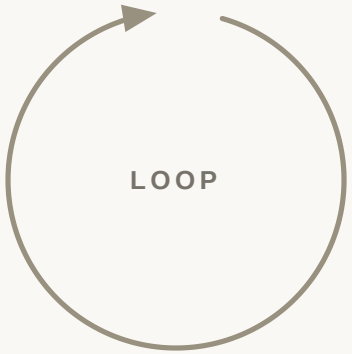
GFMI · MODEL RISK AND AI GOVERNANCE

Human in the Loop Isn't New

Nicholas J. Goble, Ph.D. · Domino Data Lab

What do we actually mean by human in the loop?



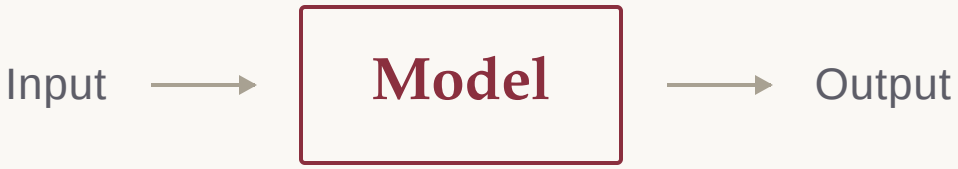
When do we add a human?

Where do we add a human?

What can we learn from how we've supervised other kinds of work?

What is a model?

For this discussion, think of a model as an operator: it takes an input and produces an output.



Human oversight helps manage the risk of using that output.

What is the human responsible for?

Taking responsibility for the decision.

For this talk, the human's role is to decide whether the result is acceptable and what happens next.

Building and using increasingly overlap.

We use models to build models.

We use agents to build agents.

Many controls assume development happens first and use comes later. With agents, those activities can happen in the same workflow.

AN EXAMPLE: AI-ASSISTED CODING

Vibe coding.

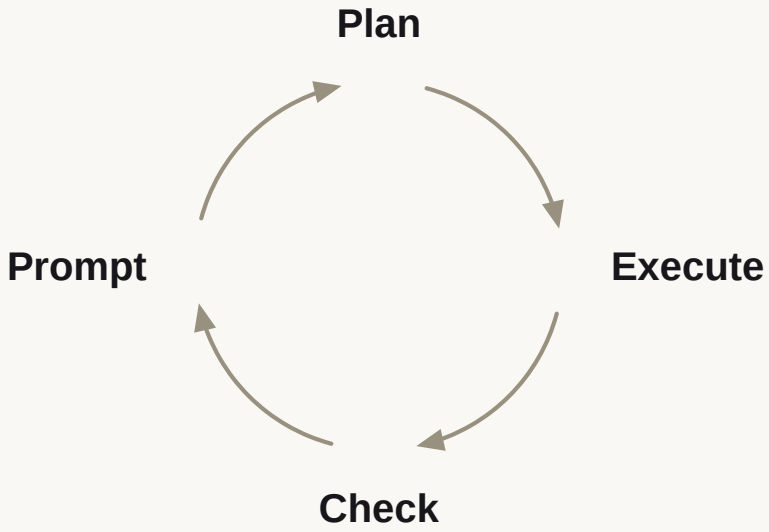
Show of hands. Who here has vibe coded?

Term coined by Andrej Karpathy, **2 February 2025**. The term is now used much more broadly.

From the outside, it looks like this.

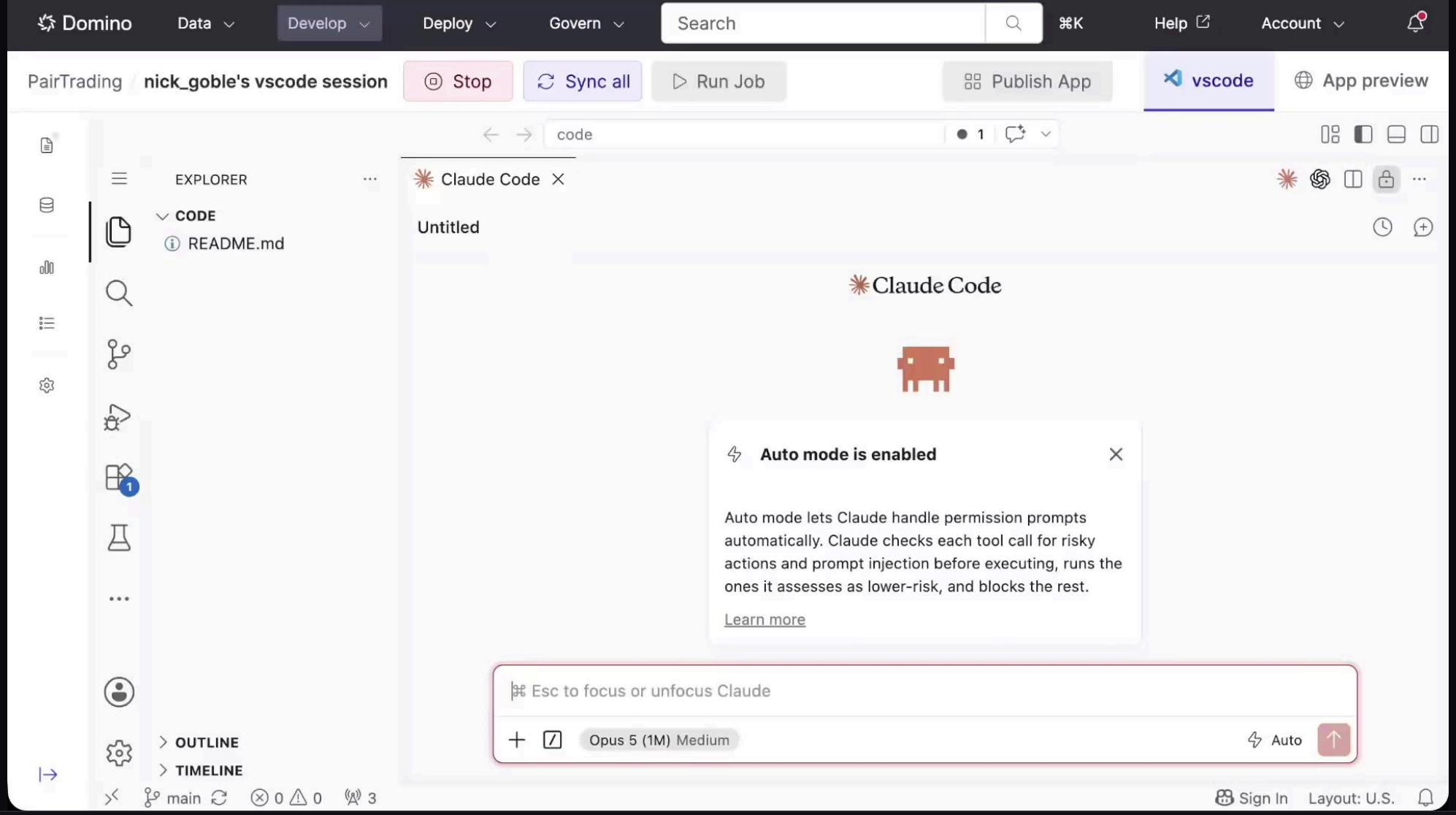


AN INFORMAL PROCESS



THE SAME WORK, AS A CYCLE

The process is iterative: prompt, plan, execute, check, repeat.

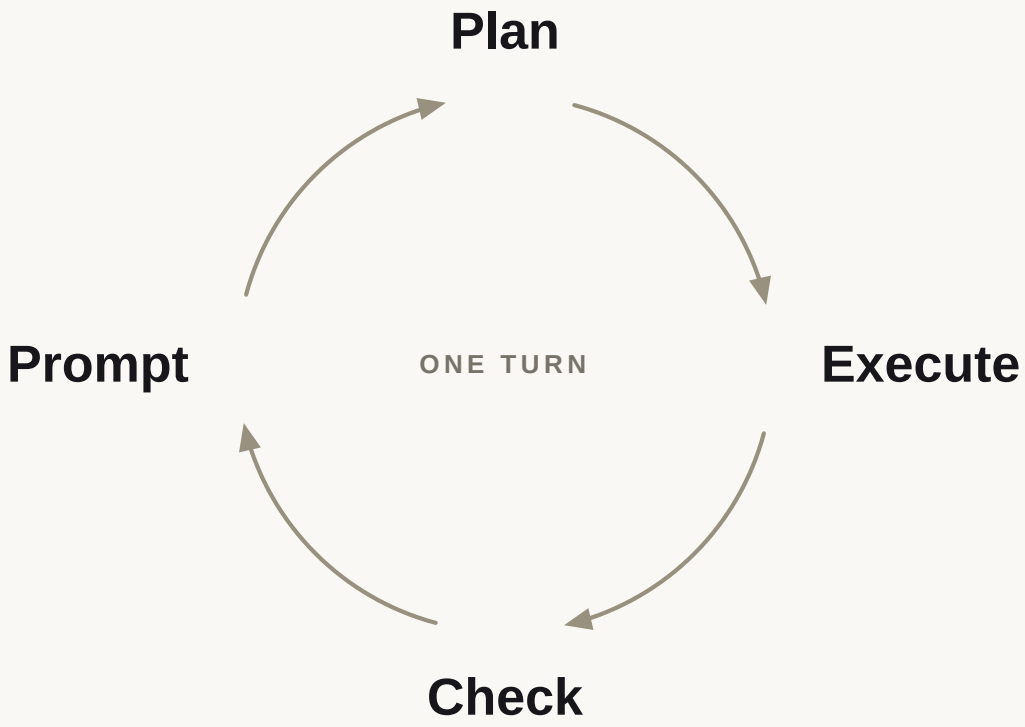


DEMO

Vibe coding, live

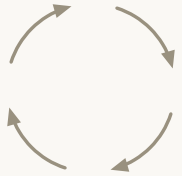
Watch how I decide whether to accept the result. Then look at what the workflow records about that decision.

The basic coding loop

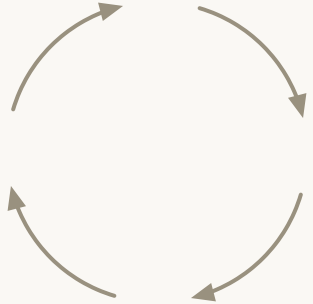


Each iteration can do more.

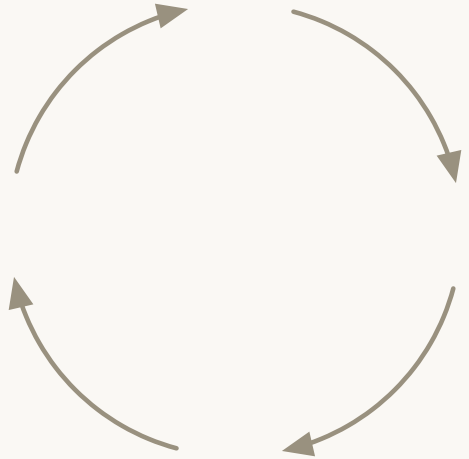
A FUNCTION



A FEATURE



A SERVICE



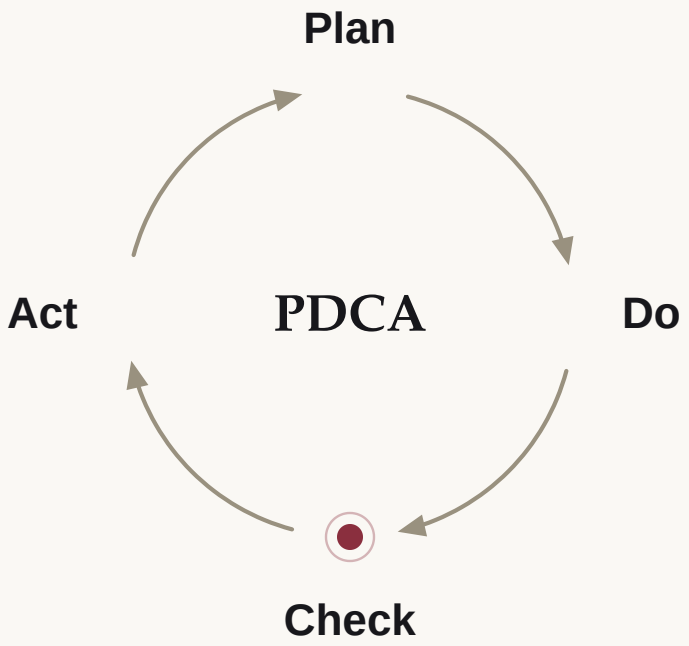
As each iteration does more, a missed error can have a larger impact. Review needs to account for the scope and risk of the change.

The loop isn't new.

We already manage iterative work through established improvement processes.

One familiar example is Plan–Do–Check–Act.

IT ALREADY HAS A NAME



PLAN

Decide what you're going to try, and what you expect to happen.

DO

Carry out the plan on a manageable scale.

CHECK

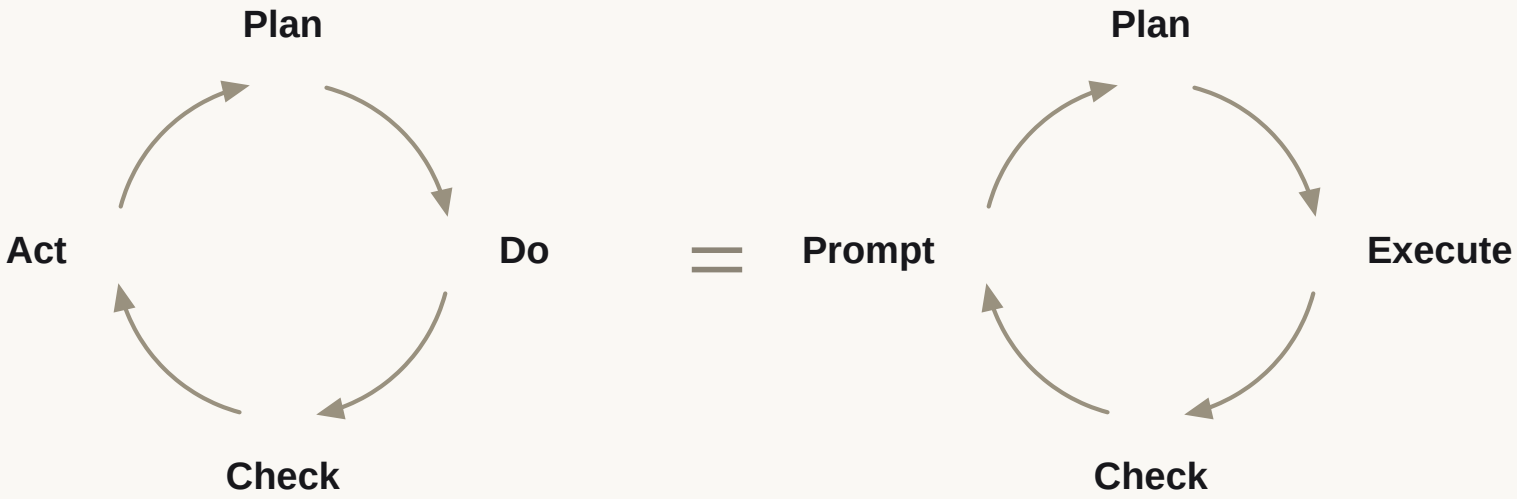
Compare what happened to what you said would happen.

ACT

Decide what to keep, change, or stop.

Traces to Walter A. Shewhart (1939), via Deming; named Plan–Do–Check–Act by Japanese practitioners in 1951. Fuller history in the speaker notes.

The coding loop follows a familiar pattern.



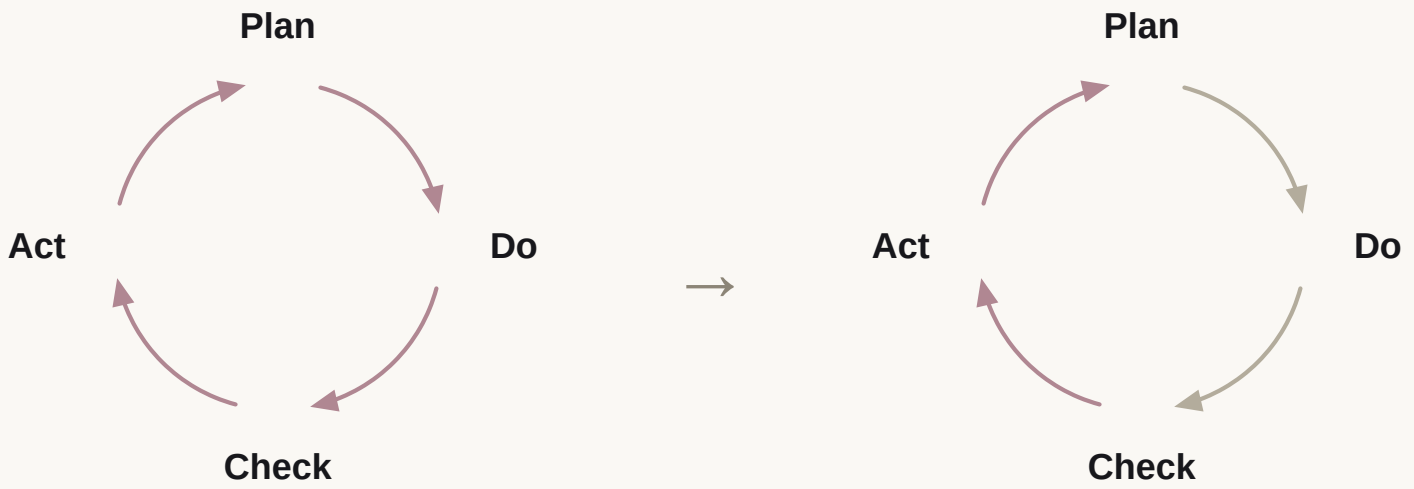
QUALITY CONTROL, 1951

YOUR ENGINEERS, THIS MORNING

Plan → Plan	**Do → Execute**	**Check → Check**	**Act → Accept, revise, or stop**

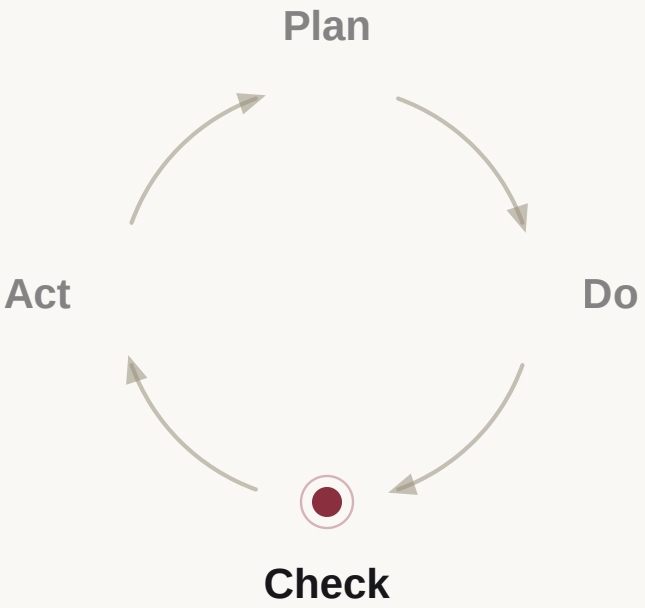
PDCA gives us a starting point for governing this work.

Agents can now perform more of the loop.



Agents can help plan, execute, and test the work. We still need to decide who is responsible for accepting the result.

Accountability has to stay in the loop. The question is where.



Check.

Check is where we assess the result against our expectations. Act is where we decide what to do next.

If acceptance is automated, a named owner still needs to be responsible for the acceptance criteria and escalation rules.

Three types of checks

FIRST LINE

CONTROL

Controls that prevent, detect, or limit errors during development and use.

Owned by the teams that build and operate the system.

SECOND LINE

VALIDATE

Independent assessment of whether the system performs as intended and is suitable for its use.

Performed with independence from development.

THIRD LINE

AUDIT

Assessment of whether governance, controls, and validation are effective.

Performed by internal audit.

So why does all of this feel new?

ONE

Some models can produce **different outputs from the same input**.

TWO

A general-purpose tool can be used for **tasks that weren't anticipated at approval**. Users may choose new tasks after the tool has been approved.

THE GUIDANCE YOU WOULD REACH FOR

> *Generative AI and agentic AI models are novel and rapidly evolving. As such, they are not within the scope of this guidance.*

SR 26-2, *Revised Guidance on Model Risk Management*, 17 April 2026; footnote 3 of the attached guidance. Interagency: Federal Reserve, OCC, FDIC. Supersedes SR 11-7 and SR 21-8. OCC counterpart Bulletin 2026-13.

The exclusion still leaves firms responsible for managing these risks through their own governance practices.

We've supervised unpredictable behavior before.

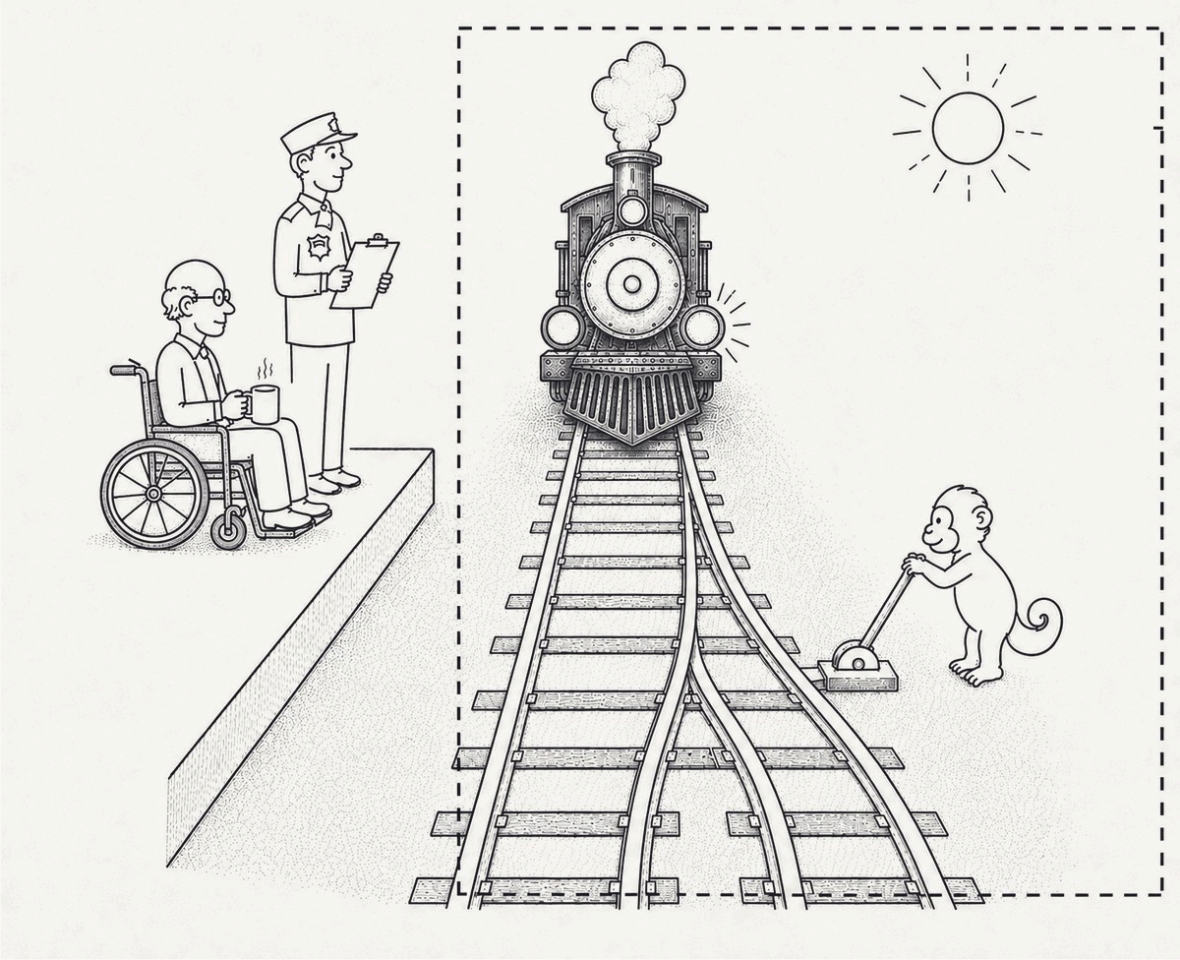
Longer than model risk management has existed as a function.

1880S · UITENHAGE, SOUTH AFRICA

Jack the baboon worked the railway signals.

James Edwin Wide lost both legs in a fall between moving railcars and came back to work as a signalman. He bought a chacma baboon at the Uitenhage market, called him Jack, and trained him to work the levers in the signal box.

According to the account, a passenger saw Jack working the signals and complained. The railway investigated.

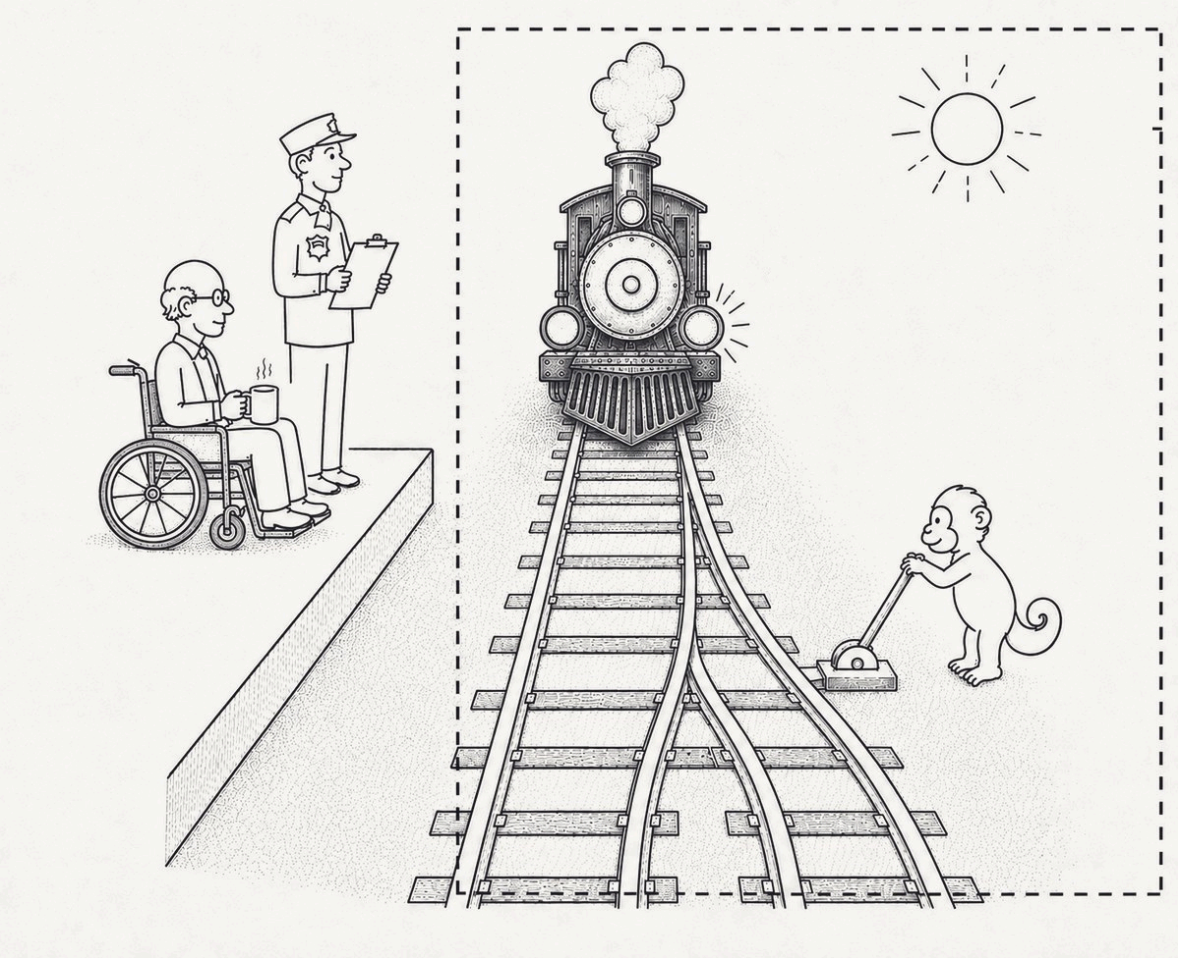


Sources: Rev. George Howe (1890) and F. W. FitzSimons, *Cape Mercury*, 29 May 1923, both cited in Nisbet, *Nature* 347:704 (1990). FitzSimons rests on an interview with Wide plus twenty-five witness statements held at the Port Elizabeth Museum. “Nine years without an error” is tradition, not record, and the inspection scene that follows is a reconstruction.

How would you supervise Jack?

You assess his performance, compare it with what should happen, and make sure someone can intervene.

APPLYING OUR THREE CHECKS TO JACK



1ST LINE **Control.** Wide supervises the work and can intervene.

2ND LINE **Validate.** An independent assessor tests whether Jack responds correctly to the signals.

3RD LINE **Audit.** Review whether supervision and testing are adequate, supported by reliable records.

Illustrative application of the framework. The historical account, and the note that the inspection scene is a reconstruction, are on the previous slide.

Build controls and evidence collection into the workflow.

Automated checks can run during execution. Independent validation and audit can use the evidence the workflow produces.

FIRST LINE

GUARDRAILS

Enforce limits on tools, data access, and actions during execution.

SECOND LINE

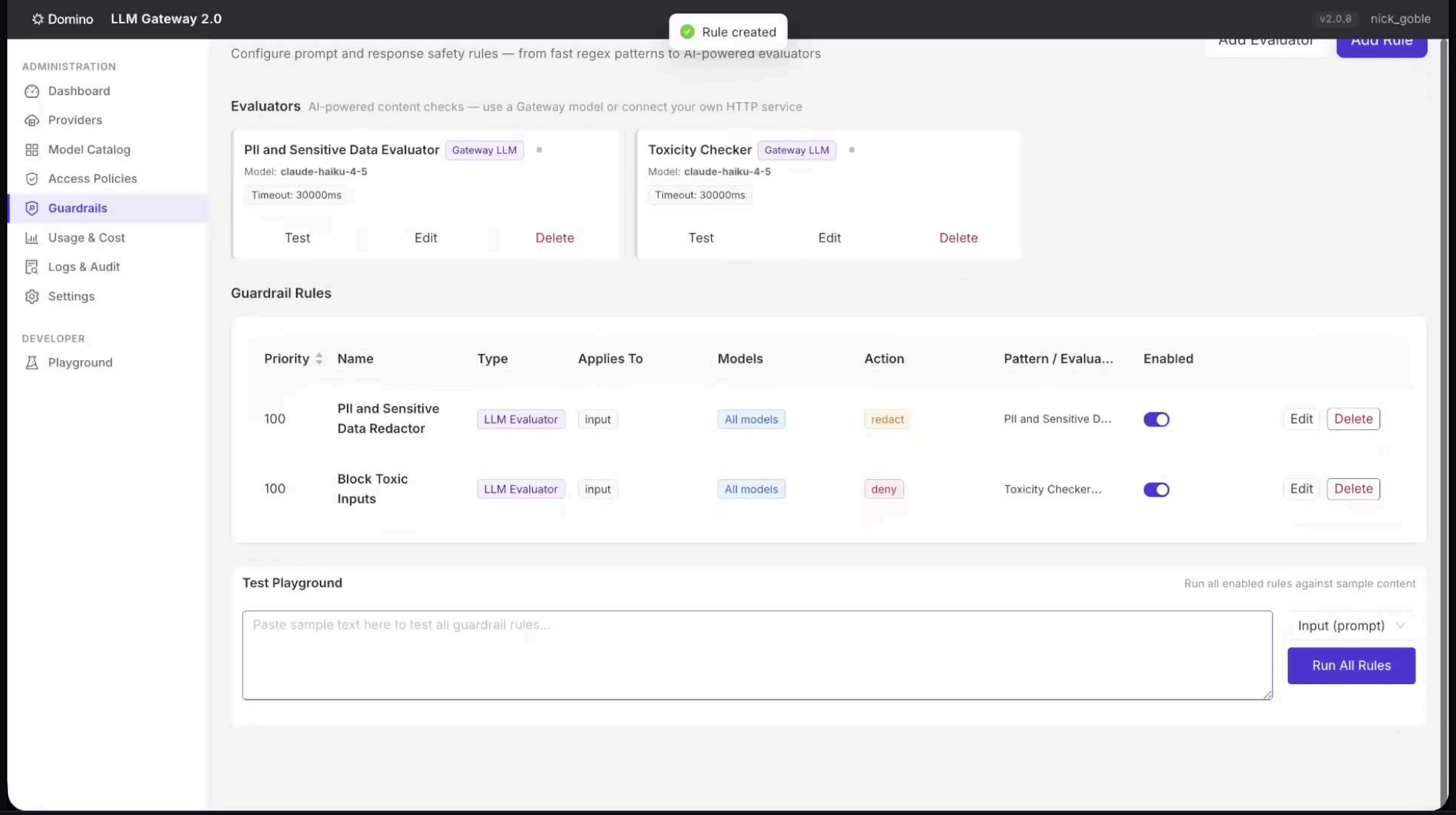
INDEPENDENT TESTS

Test against independently defined expectations, on a schedule and after material changes.

THIRD LINE

EVIDENCE FOR AUDIT

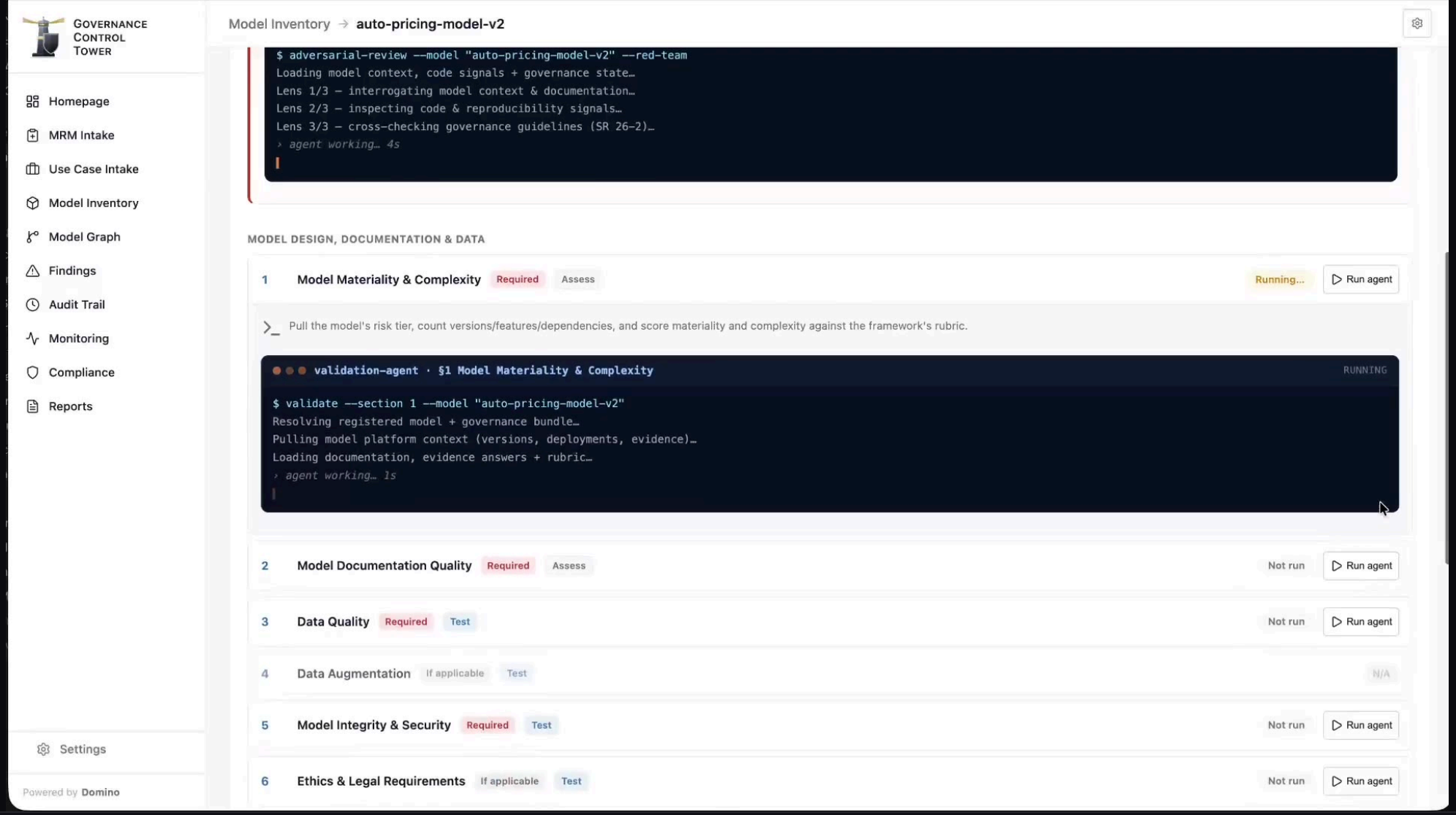
Record the inputs, actions, checks, results, and any approvals or exceptions.



DEMO · FIRST LINE

Use agents to enforce controls

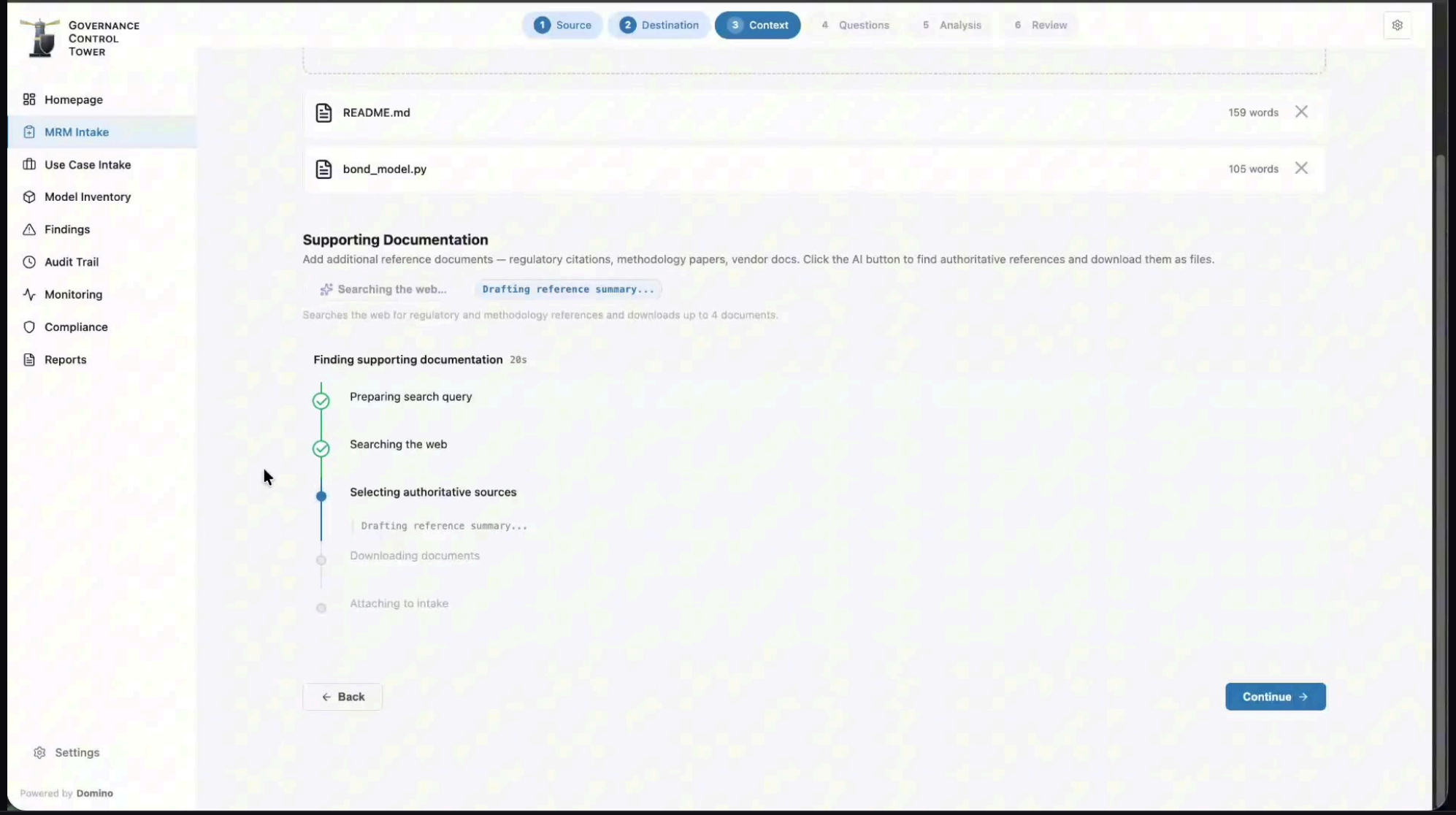
Check proposed actions against permissions and limits before execution.



DEMO · SECOND LINE

Use agents to support validation

Run tests against independently defined expectations and route findings to the validation team.

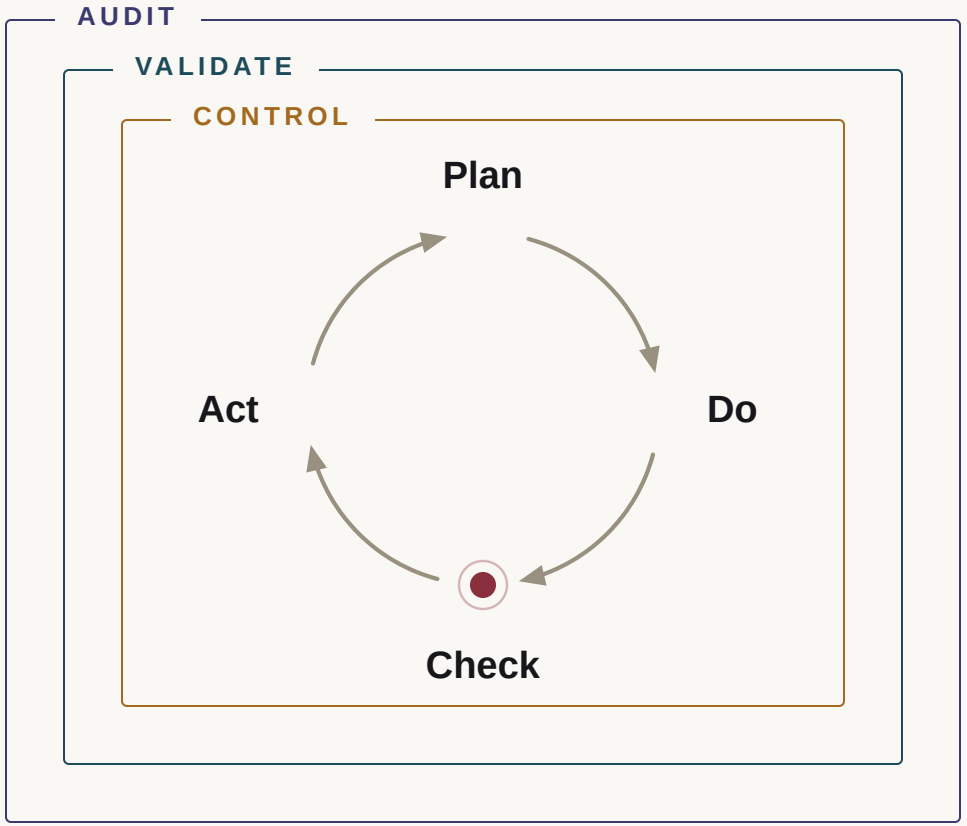


DEMO · THIRD LINE

Use agents to support audit

Collect evidence as the work happens so auditors can assess the process and investigate exceptions.

Control, validate, and audit the loop.



Build controls into execution, test independently, and retain evidence of the decisions.

BACK TO WHERE WE STARTED

When do we need human review?

When the risk requires human judgment. Define which decisions can be automated and which need review or escalation.

Where does the human fit?

At the decisions that require their judgment, with the evidence and authority to accept, reject, or change the outcome.

Define acceptance criteria and assign responsibility before the work starts. Record the decision when it happens.

Questions.

Nicholas J. Goble, Ph.D. · Domino Data Lab

The Model Operator (2026) · *The Trust Functional* (2026)